

OMER FARUK GOKDAS

Software Engineer | C/C++, Systems & AI-Augmented Engineering

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TECHNICAL SKILLS

Core Programming Languages: C, C++, TypeScript, JavaScript (ES6), SQL, Bash Shell Scripting, Java (OOP & Foundations)

Systems & Low-Level Development: Linux (Ubuntu), POSIX Unix Standards, Multithreading, Mutex Synchronization, Heap Memory Allocation, Process Control, Signals Handling

Web & Frontend Development: Astro, TypeScript, JavaScript (ES6), HTML5, CSS3, Responsive Design, Static Site Generation (SSG), DOM Manipulation

Backend & Databases: PostgreSQL, MySQL, Node.js (Express), Socket.IO, WebSockets, RESTful APIs, Relational Database Modeling

Software Engineering & Testing: Data Structures & Algorithms, Object-Oriented Programming (OOP), Debugging (GDB, Valgrind), Unit Testing, SDLC

AI-Augmented Engineering: AI Coding Agents (Antigravity, Claude, Cursor), Agentic Workflow Automation, LLM Prompt Architecture, Rapid Prototyping

Tools & Infrastructure: Git, GitHub, Version Control, Docker (Basics), Postman, Linux CLI, Active Directory, TCP/IP Networking Fundamentals

WORK EXPERIENCE

Freelance Web Developer | **Self-Employed** | Istanbul, Turkey

Jan 2023 - Present

- Architected and launched branda.ist, an 80+ page commercial platform using Astro and TypeScript, implementing automated semantic SEO architecture across 12 product categories.
- Engineered responsive component layouts and automated asset optimization with Sharp and PurgeCSS, achieving sub-1.5 second initial page load times across 80+ pages.
- Developed JavaScript (ES6) DOM validation scripts that decreased form submission errors by 30% and improved user interaction speed.

IT Intern | **Besiktas Municipality** | Istanbul, Turkey

Feb 2026 - Jun 2026

- Administered 150+ enterprise Active Directory user accounts, configuring access permissions and enforcing credential security policies.
- Collaborated across municipal departments to resolve 75+ technical support tickets across 80+ workstations, troubleshooting TCP/IP networking and OS setups.
- Developed 5 custom Bash automation scripts for routine network diagnostics, saving 3+ hours of manual system administration weekly.

PROJECTS

Minishell | **POSIX Unix Shell**

Technologies: C, POSIX System Calls, Process Control, Pipes & Redirections

- Engineered a POSIX-compliant Unix shell in C implementing 15+ built-in shell commands and runtime environment variable expansion.
- Constructed multi-pipe I/O redirection and process control architectures, achieving 100% leak-free heap execution verified by Valgrind across 50+ test suites.

Push Swap & Philosophers | **Algorithms & Concurrency**

Technologies: C, C++, Multithreading (POSIX Threads), Mutex Locks, Data Structures

- Philosophers:** Synchronized 200+ concurrent threads in C using POSIX threads and mutex locks, eradicating race conditions and preventing deadlocks.
- Push Swap:** Formulated an optimized stack-sorting algorithm in C, sorting 500 integers within 5,500 operations and reaching $O(N \log N)$ efficiency.

NishChat | **Campus Messaging Platform (Graduation Project)**

Technologies: React Native, Node.js, Express, Socket.IO, PostgreSQL

- Implemented a real-time messaging application utilizing Node.js and React Native, demonstrably reducing student communication delays by an estimated 60% via prompt message synchronization.
- Implemented bidirectional WebSocket channels via Socket.IO, achieving sub-100ms message delivery and managing 6 relational PostgreSQL tables for secure user chats.
- Validated real-time client-server synchronization across 3 physical and emulated mobile devices, maintaining zero packet loss under continuous messaging tests.

branda.ist | **Commercial Web Platform**

Technologies: Astro, TypeScript, Static Site Generation (SSG), Sharp, PurgeCSS, Programmatic SEO

- Deployed a production commercial platform (branda.ist) utilizing Astro SSG and TypeScript to render 80+ responsive catalog pages for industrial canvas products.
- Streamlined component architecture utilizing AI-assisted workflows (Claude, Cursor) to construct reusable Astro and TypeScript components, structured schema markup, and CSS styling.

EDUCATION

Software Engineering Curriculum | **42 Istanbul (Ecole 42 Network)** | Istanbul, Turkey

Jan 2025 - Present

Core Focus: C/C++, POSIX Unix Standards, Multithreading, Heap Memory Allocation; Conducted 40+ peer code evaluations and collaborative reviews.

Bachelor of Science in Software Engineering | **Nisantasi University** | Istanbul, Turkey

Sep 2021 - Jul 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Object-Oriented Programming, Computer Networks

CERTIFICATIONS AND AWARDS

- 3rd Place Award:** National Informatics Valley Web Hackathon (2024).
- Green Software Practitioner Certification:** Green Software Foundation (2025).
- Software Technologies & AI Program Certification:** GEN Academy (2024).
- C Programming Language Certification:** BTK Academy (2023).

LANGUAGES

- Turkish:** Native | **English:** Professional Working Proficiency (B2).